

More on groups and rings.

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Recall that a group  $G$  is a set together with an associative operation,

identity, and inverses. A ring is

an abelian with multiplication. We

now give more details in their studies.

Def 1: let  $G$  be a group, and

$\emptyset \neq H \subseteq G$  a subset. We say  $H$  is a subgroup

of  $G$  if  $H$  is a group with the group

operation of  $G$  restricted to  $H$ .

eg.  $2\mathbb{Z} = \{2n : n \in \mathbb{Z}\} \subseteq (\mathbb{Z}, 0, +)$

is a subgroup.

Is  $2\mathbb{Z}+1 = \{2n+1 : n \in \mathbb{Z}\}$  a subgroup?

Lemma 2: Let  $G$  be a group and

$H \subseteq G$  a subset. Then  $H$  is a subgroup

iff (1) identity  $e_G \in H$

(2) if  $h_1, h_2 \in H$ , then  $h_1 h_2 \in H$

(3) if  $h \in H$ , then  $h^{-1} \in H$ .

proof: (1)  $\Rightarrow$  (2) and (3) is required

by the definition of group. We show

(ii) holds by showing  $e_H = e_G$ .

Note  $e_H e_H = e_H$

So multiplying  $e_H^{-1}$ , we get

$$e_H^{-1} e_H e_H = e_H^{-1} e_H$$

$$e_H = e_G e_H = e_G$$

( $\Leftarrow$ ) obvious, by setting  $e_H = e_G$ .

eg.  $(\mathbb{Z}_6, +)$  has a subgroup

$$\{[0], [2], [4]\} = [2] \mathbb{Z}_6$$

$(\mathbb{Z}, +)$  has a subgroup

$$m\mathbb{Z} = \{m \cdot k : k \in \mathbb{Z}\} \quad \forall m \in \mathbb{Z}$$

Def 3: let  $G$  be a group and

$g \in G$ . Define  $\langle g \rangle = \{ g^n : n \in \mathbb{Z} \}$ .

Lemma 4:  $\langle g \rangle$  is a subgroup of  $G$ .

This is called the cyclic group generated

by  $g$ . eg.  $m\mathbb{Z}$ ,  $[2]\mathbb{Z}_6$ ,  $\mathbb{Z}$ .

Def 5. let  $G$  be a group and  $g \in G$ .

(1) The order of  $G$  is  $|G|$

(2) The order of  $g$  is

$$|g| = \min \{ d > 0 : g^d = e_G \}.$$

eg. Consider  $(\mathbb{Z}_6, [0], +)$  has order 6.

$$[2] + [2] + [2] = [6] = [0]$$

$$[3] + [3] = [6] = [0].$$

$$\text{So } |[2]| = 3 \text{ and } |[3]| = 2.$$

Def 6. Let  $R$  be a ring and  $\emptyset \neq S \subseteq R$

be a subset. We say  $S$  is a subring

if  $S$  is a ring with ring operations

of  $R$  restricted to  $S$ .

eg.  $\mathbb{Z}$  is not a subring of  $\mathbb{Q}$

Lemma T: let  $R$  be a ring and  $S \leq R$ .

Then  $S$  is a unital subring if

$$(1) \quad 1_R \in S$$

$$(2) \quad \forall x, y \in S. \quad x - y \in S$$

$$(3) \quad \forall x, y \in S. \quad xy \in S$$

iff  $S$  is a subgroup (of  $R$  viewed as  
an abelian group) that is closed

under multiplication.

eg. Can you find a subring of  $\mathbb{Z}$

is not of the form  $m\mathbb{Z}$  for some  $m \in \mathbb{Z}$ ?

Note a subring of  $\mathbb{Z}$  is automatically a subgroup of  $\mathbb{Z}$ . So we should also answer the question is there a subgroup of  $\mathbb{Z}$  that is not of the form  $m\mathbb{Z}$ .

Prop. 8. All subgroups of  $\mathbb{Z}$  is of the form of  $m\mathbb{Z}$  for some  $m \geq 0$ .

proof. Let  $H \subseteq \mathbb{Z}$  be a subgroup.

let  $m$  be the least positive element in  $H$ . If  $m$  does not exist.

then  $H = \{0\} = 0\mathbb{Z}$ .

If  $m$  exists, we show for  $x \in H$ ,

$$x = qm \quad \text{for some } q \in \mathbb{Z}.$$

By the division algorithm,

$$x = qm + r \quad \text{for } 0 \leq r < m.$$

If  $r \neq 0$ , then  $r = x - qm$

$$= x - \underbrace{m - \dots - m}_{q \text{ copies of } m} \in H \quad (\text{why?})$$

But  $0 < r < m$  contradicts the

minimality of  $m$ . So  $r = 0$ .

This implies  $H \subseteq m\mathbb{Z}$  because all elements in  $H$  is divisible by  $m$ .

Conversely, since  $m \in H$ , and  $H$  is closed under addition and inverse.

$m\mathbb{Z} \subseteq H$ . Therefore  $H = m\mathbb{Z}$ .

Prop 9: Any unital subring of  $\mathbb{Z}$  must of  $\mathbb{Z}$ .

proof: let  $S \subseteq \mathbb{Z}$  be a subring.

Then  $1 \in S$ . Hence  $\mathbb{Z} = \langle 1 \rangle \subseteq S$ .